

$\llbracket _ \rrbracket^c : \text{Com} \rightarrow \mathcal{S}_c$ is defined as:

$$\llbracket x ::= E \rrbracket^c(s) \triangleq sc_{asg}(\llbracket E \rrbracket^e, x)(s)$$

$$\llbracket \emptyset \rrbracket^c(s) \triangleq s$$

$$\llbracket C_1; C_2 \rrbracket^c(s) \triangleq st_{seq}(\llbracket C_1 \rrbracket^c, \llbracket C_2 \rrbracket^c)(s)$$

$$\llbracket B ? C_1 : C_2 \rrbracket^c(s) \triangleq st_{cnd}(\llbracket B \rrbracket^b, \llbracket C_1 \rrbracket^c, \llbracket C_2 \rrbracket^c)(s)$$

$$\llbracket \uparrow B : C \rrbracket^c(s) \triangleq (\llbracket B \rrbracket^b, \llbracket C \rrbracket^c)(s)$$

Assignment $sc_{asg}(e, x)(s) =$

$$\begin{cases} s[x \mapsto e(s)] & e(s) \downarrow \\ \perp & \text{o.w.} \end{cases}$$

Sequencing $sc_{seq}(c_1, c_2)(s) =$

$$\begin{cases} c_2(c_1(s)) & c_1(s) \downarrow \\ \perp & \text{o.w.} \end{cases}$$

Conditional $sc_{cnd}(b, c_1, c_2)(s) =$

$$\begin{cases} c_1(s) & b(s) \downarrow \wedge b(s) = \text{tt} \\ c_2(s) & b(s) \downarrow \wedge b(s) = \text{ff} \\ \perp & \text{o.w.} \end{cases}$$

$st_{whl}(b, c) = st_{cnd}(b, st_{seq}(c, st_{whl}(b, c)), st_{skp})$, but this is not a definition. Instead:

Assume $F : (A \rightarrow A_{\perp}) \rightarrow (A \rightarrow A_{\perp})$:

1. Define $seq \forall i \in \mathbb{N}. f_i : A \rightarrow A_{\perp}$ where $f_0 \triangleq \perp$ and $f_{i+1} \triangleq F(f_i)$.
2. Define func $f_{\infty} : A \rightarrow A_{\perp}$ st: $f_{\infty}(a) \triangleq \begin{cases} a' & \exists i. f_i(a) = a' \neq \perp \\ \perp & \text{o.w.} \end{cases}$
3. Ass $\forall i, a. [f_i(a) \downarrow \Rightarrow f_{i+1}(a) = f_i(a)]$.

By **fixpoint theorem**, f_{∞} is a fixpoint of F . For any other fixpoint g of F , $\forall a. [f_{\infty}(a) \downarrow \Rightarrow f_{\infty}(a) = g(a)]$.